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Atomic diffusion behavior near the bond interface during the explosive welding process based on molecular dynamics simulations

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ABSTRACT

Based on the Hastelloy/stainless steel explosive-welded composite (the main elements are Ni and Fe respectively), this work investigated the diffusion behavior of main elements near the bond interface during the explosive welding process through molecular dynamics simulations and experiments. According to the characteristics of temperature and pressure of explosive welding process, the simulation of explosive welding process is classified into three stages, namely loading stage, unloading stage, and cooling stage. The diffusion behavior of the main elements (Ni and Fe) near the interface was investigated by two methods. The first one is to calculate the mean square displacement of the main elements. And the second one is to compare the system configurations and the results of radial distribution function at different stages. In the simulation, the thickness of the diffusion layer of the V_{px} (the transverse velocity)= 0.2 km/s, V_{px}= 0.4 km/s, V_{px}= 0.6 km/s, V_{px}= 0.8 km/s weldments are about 14, 20, 49, and 61 Å. The results show that the atomic diffusion near the interface mainly occurs in the unloading stage and the beginning of the cooling stage. Only when the melting occurs during these stages, the atoms of the bonding interface can diffuse obviously. In this experiment, the thicknesses of the diffusion layer at the wave peak and the vortex zone are about 20 μm and 230 μm respectively. This conclusion is verified by the results of experimental investigations and the references. Also, the diffusion layer thickness with the increasing the absolute value of V_{pz} (the longitudinal velocity) and V_{px}, and the absolute value of V_{pz} can make a greater contribution to the thickness of the diffusion layer than that of V_{px}. The thickness of the diffusion layer at the interface will be significantly increased if the melting occurred during the welding process.

1. Introduction

As a solid-state welding method [1–5], explosive welding is an effective technology to produce composite plates without producing an obvious heat-affected zone [6,7]. Thus, it is one of the main methods for welding similar or dissimilar materials currently. Fig. 1(a) is a schematic diagram of the explosive welding. During this welding process, the explosive is used to drive the flyer plate to impact the base plate at a high speed, so the flyer plate and the base plate can be combined instantly [8–10]. Due to the high strain rate and the severe plastic deformation [11,12], this explosive welding process is difficult to study directly. Numerical simulation technology has become an effective method to study the process of explosive welding because of its advantages, such as low calculation cost and the ability to capture the transient phenomena in the explosive welding process and so on [13]. At present, the mainstream simulation methods used in explosive welding are as follows: Eulerian, Lagrangian, arbitrary Lagrangian-Eulerian (ALE), and smoothed particle dynamics method (SPH) [14]. Reference [11] used the SPH method to explore the temperature distribution near the interface of a titanium/steel explosive-welded composite and explained the size difference of interfacial grain by recovery-recrystallization-recrystallization (growth) mechanism. Reference [15] used the Lagrangian/Euler coupling model to explore the optimal welding parameter and experimented with this parameter to obtain a well-combined Al/Fe explosive-welded composite. Reference [4] used SPH and ALE methods to study the welding process of the titanium/aluminum explosive-welded composite; the waveform interface of the simulation result has a high consistency with the interface morphology observed in the experiment.

However, the range scale of the simulation is usually from a few millimeters to a few hundred millimeters by the above methods [4,11, 15,16]. It is hard to explore the atoms' diffusion behavior near the

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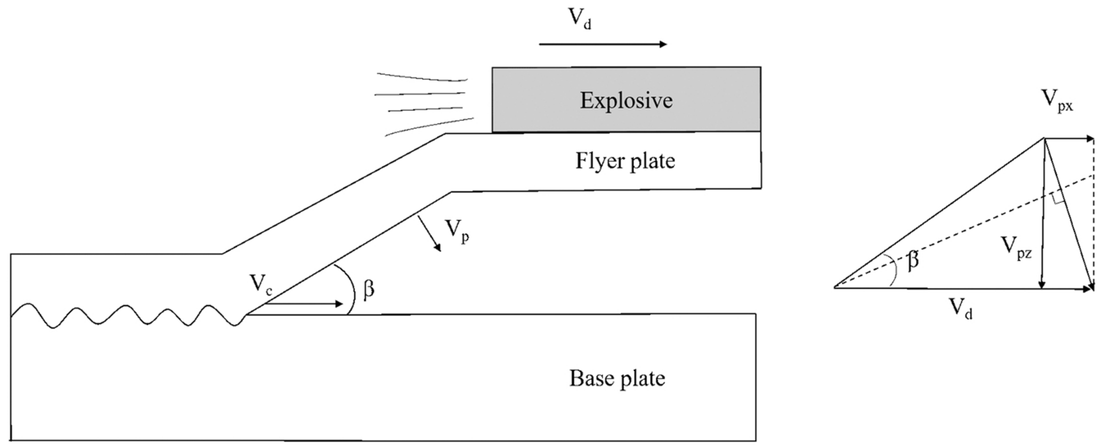

Fig. 1. (a) Schematic diagram of explosive welding process: V_d is the detonation velocity, V_c is the collision point velocity, V_p is the velocity of flyer plate, β is the collision angle (b) geometric analysis of explosive-welded parameters.

Table 1
Chemical composition of plates used in the experiment.

	Mo	Cr	Fe	W	Co	C	Si	Mn
C-276	15.0–17.0	14.5–16.5	4.0–7.0	3.0–4.5	≤ 2.5	≤ 0.01	≤ 0.08	≤ 1.0
304 L	—	18.0–20.0	Bal.	—	—	≤ 0.03	≤ 0.75	≤ 2.0
	V	**Ni**	**S**	**P**				
C-276	≤ 0.35	Bal.	≤ 0.03	≤ 0.04				
304 L	—	8.00–12.00	≤ 0.02	≤ 0.04				

interface during explosive welding. Diffusion, especially the atomic inter-diffusion near the interface of dissimilar metals, is generally considered to be a crucial bonding mechanism for joining dissimilar metals [17]. The thickness and diffusion behavior of the diffusion layer directly or indirectly affect the bonding properties and strength of the composites [9]. Therefore, the study of atomic diffusion behavior in explosive welding is of great significance to understand the welding process and welding mechanism, and even provides an opportunity to improve the bonding strength of composite materials. Also, it is always a controversial topic whether there is obvious atomic inter-diffusion near the interface during explosive welding. Reference [18] used an energy dispersive spectrometer (EDS) to analyze the element distribution near the interface of the aluminum alloy/magnesium alloy explosive-welded composite and found an element diffusion layer about 3 μm thick at the peak and valley of the wavy interface. Reference [19] used EDS to analyze the element distribution near the interface of a Cu/steel explosive-welded composite, and the results showed that there was almost no element inter-diffusion on both sides of the interface. Reference [20] used scanning electron microscope (SEM), EDS and transmission electron microscope (TEM) to analyze no intermetallic compound or second phase is formed at the composite interface of SKD61 and Cu after explosive welding and heat treatment.

The current research on the atomic diffusion behavior in the explosive welding process are mainly focused on experiments, and the scope of research is also on the micron level. At the atomic scale, there are only a few reports on the simulation of the atomic diffusion behavior during the explosive welding process. Molecular dynamics (MD) simulation can reproduce the entire explosive welding process at the atomic scale, so it is particularly suitable for investigating the diffusion behavior of elements during the welding process. Reference [21] established a molecular dynamics model to predict the self-diffusivity of Ni-XCr alloys with different Cr concentrations at high temperatures. Reference [9] simulated the explosive welding process of Cu/Al composites by a molecular dynamics model and explored the atomic diffusion behavior near the interface. And the thickness of the diffusion layer was estimated, which was consistent with the experimental result. Reference [22] reproduced the welding process of an Al/Mg explosive-welded composite by MD simulation and analyzed the relationship between welding parameters and atomic diffusion behavior. Reference [23] analyzed the interface morphology and atomic structure of copper and iron explosive welding at three stages by molecular dynamics. Reference [24] used molecular dynamics to model explosive welding, and the results showed that a flat interface is formed in solid-state welding while a wavy interface is produced in liquid-state welding.

Hastelloy (a nickel-based alloy), because of its high resistance to temperature and corrosion, can be widely used in various industries, such as petrochemicals, marine engineering, and the nuclear industry [25,26]. But the material cost of Hastelloy limits its application. In practice, Hastelloy/steel composites are commonly used as a substitute for Hastelloy. [27].

To further explore the atomic diffusion behavior near the interface during the explosive welding process, this paper takes Hastelloy alloy/stainless steel (metal grade is C276/304 L) explosive composite plate as the research object. Through molecular dynamics theory and experimental observation, the diffusion behavior of the main elements (Ni and Fe) atoms at the explosive composite interface during explosive welding was investigated in detail.

Based on the main elements (Ni and Fe) at the interface of different stages, the mean square displacement, the model system structure, and the radial distribution function were compared and analyzed to study the rules of atomic diffusion at the interface of the composite plate. Especially the factors affecting the atomic diffusion were discussed, for example, the welding process parameters, whether melting occurs during welding, and so on. The simulation results were verified by experimental observation. These findings will be helpful to the optimization of welding parameters in explosive welding experiments.

2. Simulation procedure

In this study, the large-scale atomic/molecular massively parallel simulator (LAMMPS) is used to reproduce the welding process of C276/304 steel (The main elements of the two materials are shown in Table 1)

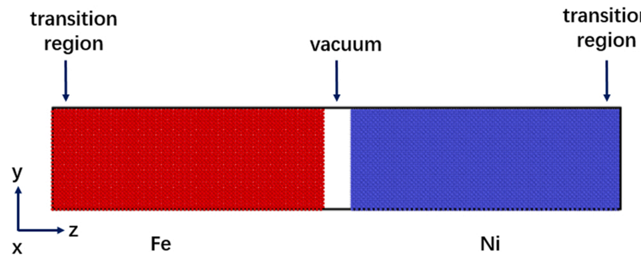

Fig. 2. Initial model structure for MD simulation. The red parts are single-crystal Fe (BCC structure) and the blue parts are single-crystal Ni (FCC structure).

explosive-welded composite [28]. In the MD simulation, the interactions of Fe and Ni atoms are described by the embedded atom method potential developed by Bonny G et al. [29]. Using this potential, the lattice constants of Fe (body-centered cubic (BCC) structure) and Ni (face-centered cubic (FCC) structure) are determined to be 2.85 Å and 3.52 Å at 300 K, which is in good agreement with the actual lattice constants. Fig. 2 shows the initial model structure for MD simulation. Single-crystal Fe (BCC structure) and single-crystal Ni (FCC structure) are placed on both sides of the simulation box along the z-direction. The vacuum (20 Å thickness) is placed in the center of the model to avoid interaction between Ni and Fe atoms during the relaxation stage. The contact surfaces of Fe and Ni are both (0 0 1) planes, and the simulation system has 193, 424 atoms. The dimensions of Fe and Ni are about 74.1 Å (x) × 74.1 Å (y) × 199.5 Å (z) and 73.92 Å (x) × 73.92 Å (y) × 197.1 Å(z) respectively, and the mismatch on the (0 0 1) plane is about 0.24%. On both sides of the simulation box, two transition regions with 3 layers of atoms are set up to reduce the adverse effect of shock waves [9]. The whole MD simulation uses the fixed time step (The time step is fixed to 1 fs throughout the simulation). Initial thermal velocities of all atoms in this system are assumed to obey the Maxwell distribution. Periodic boundary conditions are applied in the x and y directions. The initial system is relaxed for 1000 ps at 300 K and zeros external pressure under a constant temperature and constant pressure ensemble (NPT) to obtain a stable initial model.

Due to the stress changes of composite material during the collision process, the explosive welding process can be classified into the loading stage and the unloading stage[9,30]. At the same time, the temperature of the composite material usually reaches a very high value during the explosive process, even exceeding the melting point of the welding material [31]. Therefore, the influence of internal temperature on the atomic diffusion behavior should be considered during the welding process. In this study, the MD simulation of the explosive welding process was carried out in three successive stages: the loading stage, the unloading stage, and the cooling stage. Fig. 3 shows the variations of temperature and pressure during the explosive welding process. After relaxing the initial model, the transition region of the Fe bulk is fixed and the Ni bulk is given the initial speed to simulate the collision process. During the simulation, the Ni bulk moves to the Fe bulk, when the thickness of the Ni bulk reaches the shortest length, the transition region of Ni will be fixed [9,32]. After that, the system is relaxed for 1000 ps under the micro-canonical (NVE) ensemble to finish the loading stage of explosive welding. Due to the collision process, the temperature and pressure of the simulation system rise to high values during the loading stage. At the final equilibrium temperature of the loading stage, the system is relaxed for another 1000 ps at zero external pressure under the NPT ensemble to simulate the unloading stage. Finally, the cooling process with a temperature from the equilibrium temperature of the unloading stage to 300 K is simulated. Limited by the simulation time scale, the cooling rate in this article is set to 10^{12} K/s. The system will be relaxed for another 1000 ps at 300 K and there will be zero external pressure under the NPT ensemble after the system temperature drops to 300 K.

In the actual welding process, explosive-welded parameters may have a great influence on the bonding performance of composite materials. To obtain an ideal composite plate, it is usually necessary to control the welding parameters, such as the denotation velocity V_d, the collision point speed V_c, the flyer plate speed V_p and the collision angle β. There is a geometric relationship between these parameters. As shown in Fig. 1(a), when the parallel assembly method is adopted to weld the composite material, the welding parameters will satisfy the following equation.

$$V_c = V_d \tag{1}$$

$$V_p = 2V_d sin\frac{\beta}{2} \tag{2}$$

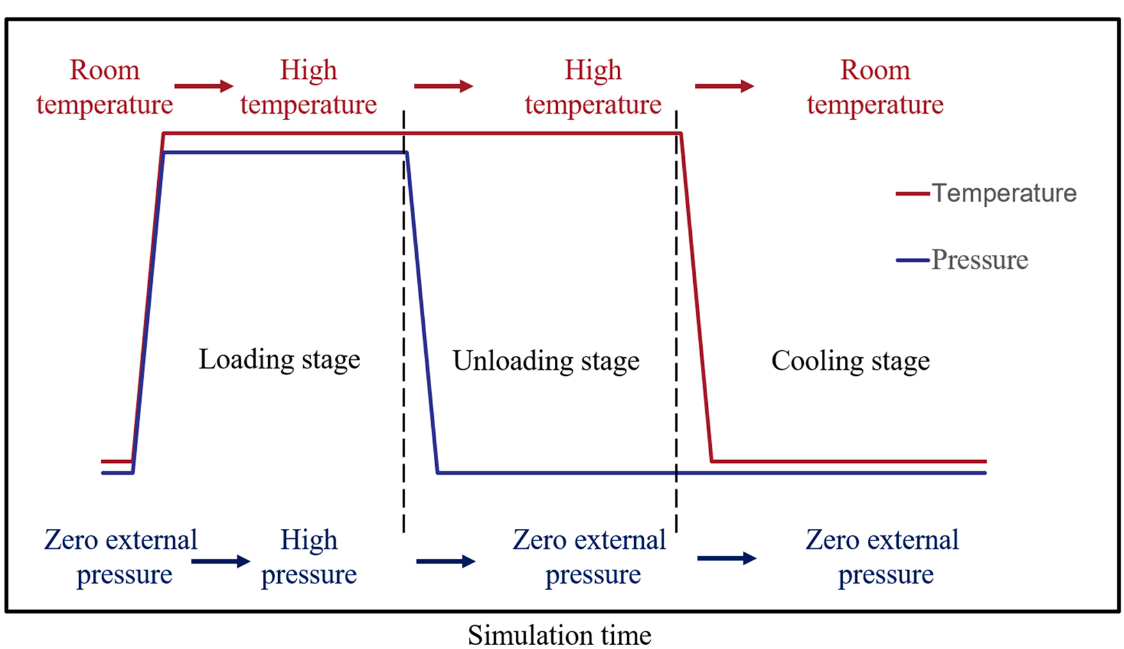

Fig. 3. Temperature and pressure changes during the explosive welding process.

Table 2
Initial welding parameters for MD simulation.

No.	V_p (km/s)	V_{pz} (km/s)	V_{px} (km/s)
1	1.61	-1.6	0.2
2	1.65	-1.6	0.4
3	1.71	-1.6	0.6
4	1.79	-1.6	0.8

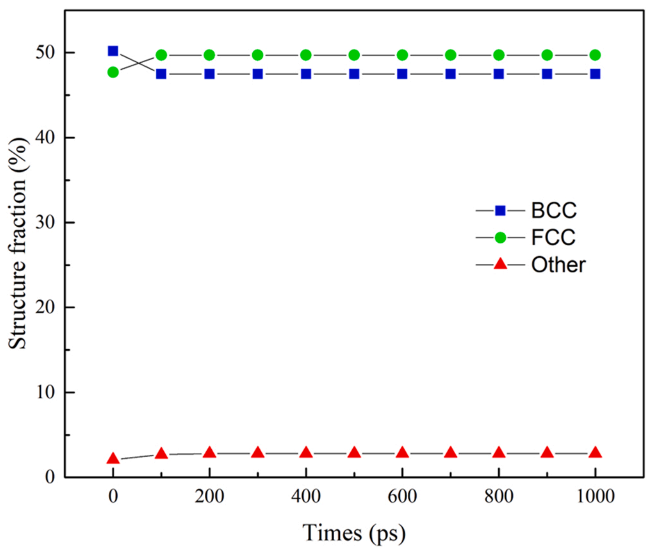

Fig. 4. Structure fraction of Ni and Fe atoms during the relaxation process.

According to the geometric analysis of V_p in Fig. 1(b), the V_p can be divided into two components: V_{pz} (the longitudinal velocity) and V_{px} (the transverse velocity). They should satisfy the following equation [22]:

$$V_{pz} = -V_d sin\beta \tag{3}$$

$$V_{px} = 2V_d sin^2\frac{\beta}{2} \tag{4}$$

Walsh et al. believed that the V_c should be smaller than the bulk sound velocity of the composite materials [33], while the bulk sound velocity of commonly used engineering materials is about 4.5–6.5 km/s [6]. Considering the actual welding parameters in the previous articles [8,11,14] and the welding parameters used in MD simulations [9,22, 23], the initial welding parameters for MD simulation in this article are shown in Table 2. The V_{pz} is fixed at -1.6 km/s, and the V_{px} changes from 0.2 to 0.8 km/s with an increase of every 0.2 km/s, so that the relationship between V_{px} and atomic diffusion behavior can be investigated first.

By defining the interfacial region where the concentration of Ni and Fe atoms both exceeds 5% as the diffusion layer, the z-length of the diffusion layer can be calculated.

To investigate the atomic diffusion behavior at different stages, the “compute-msd” command of the LAMMPS package is used to calculate the mean square displacement (MSD) of the Ni and Fe atoms during the welding process. There is a certain correlation between MSD and the diffusion coefficient [34], and the diffusion coefficient is generally regarded as an important parameter to describe the atomic diffusion behavior [35,36]. The MSD calculation formula is as follows:

$$MSD = < \sum_{1}^{n} [r_i(t) - r_i(0)]^2 > \tag{5}$$

Where $r_i(t)$ is the position of the particle i at time t, and the angle brackets "< >" is the mean value of square displacement of all particles.

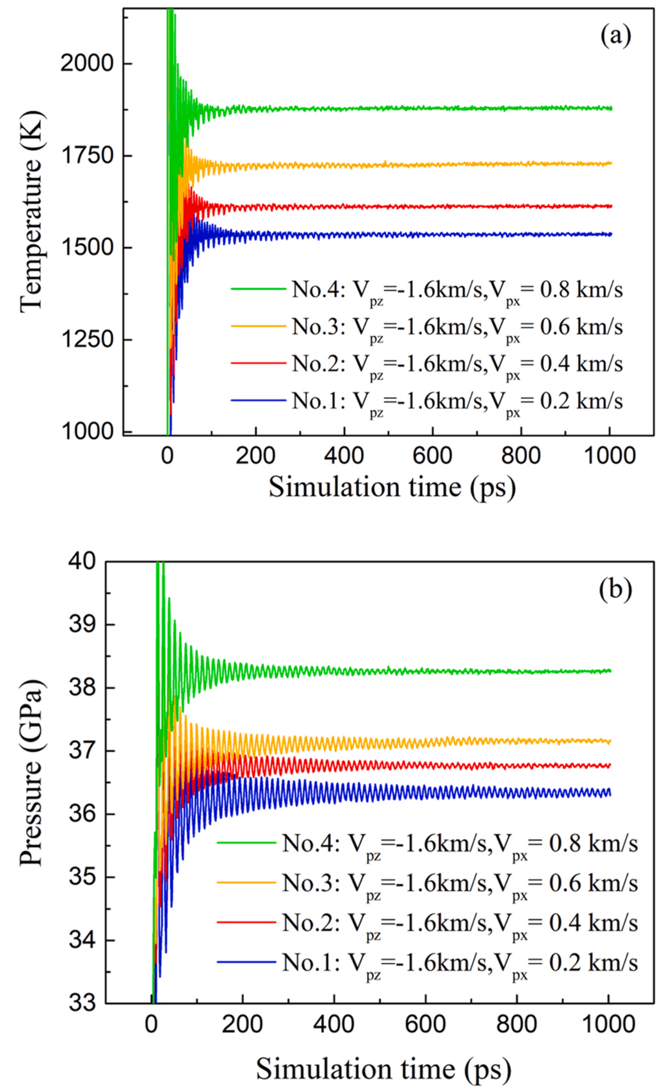

Fig. 5. Temperature (a) and pressure (b) curves of simulation system during the loading stage.

The formula for calculating the diffusion coefficient (D) is as follows:

$$D = \lim_{t\to\infty}\frac{1}{2N} < \sum_{1}^{n} [r_i(t) - r_i(0)]^2 > \tag{6}$$

Where N is the dimension of the system. If the system configuration in this paper is considered to be three-dimensional and take $N = 3$ and 1/6 of the slope of the MSD curve is the atomic diffusion coefficient. By defining the interfacial region where the concentration of Ni and Fe atoms both exceeds 5% as the diffusion layer. If the element of Ni and Fe atoms on both sides is less than 5%, it is considered to be the thermal vibration of atoms in the metal system. At this time, there may be very few atoms for diffusion migration, and this part of diffusion is ignored in this paper. If the MSD value increases with the simulation time, it indicates that the atoms are diffusing in the system. At the same time, the slope of the MSD curve is proportional to the atomic diffusion coefficient. On the contrary, if the MSD value is maintained at a certain value over the simulation time, it indicates that the type of atoms does not diffuse during this time [9,22].

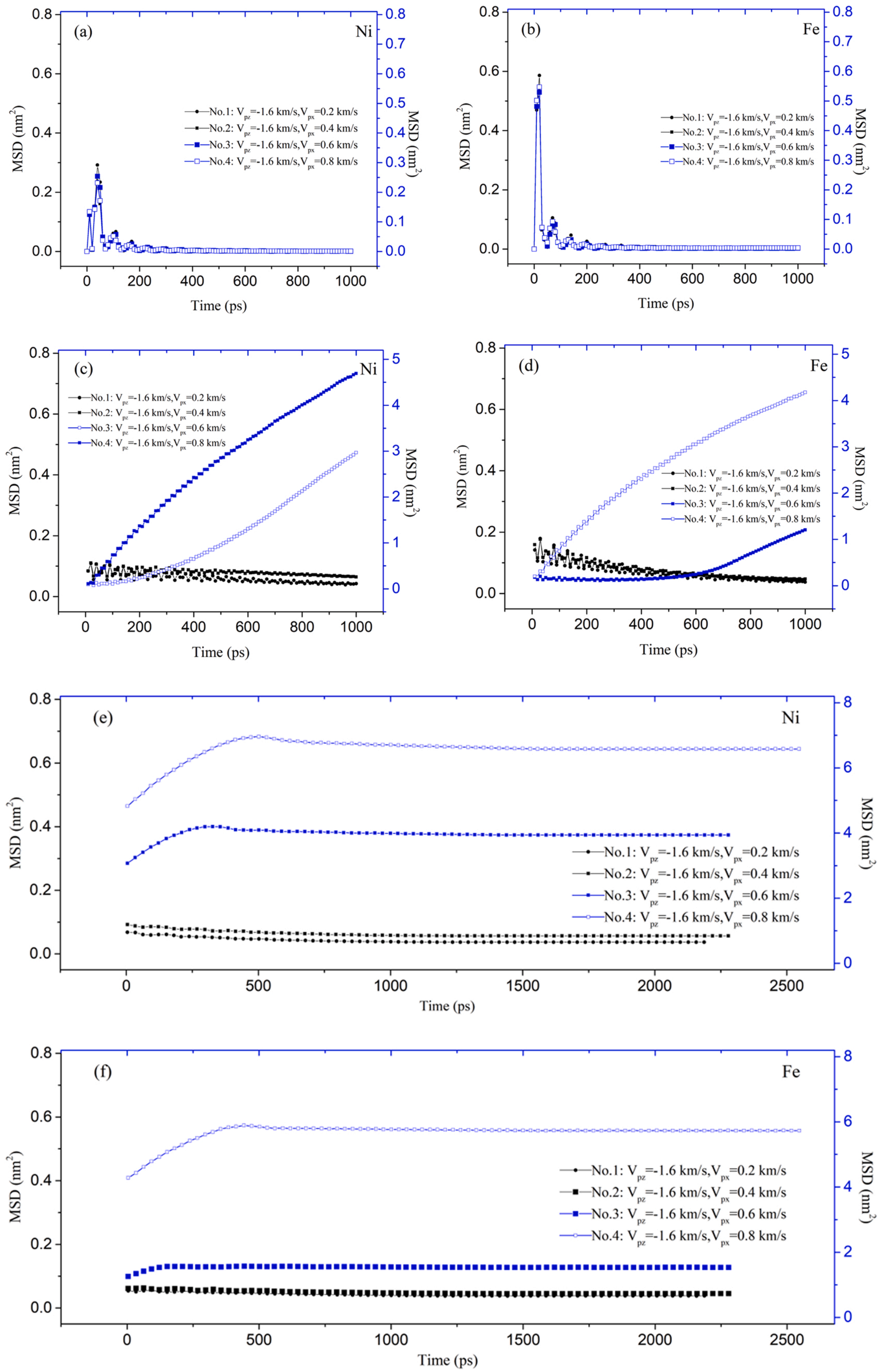

Fig. 6. MSD curves of Ni and Fe atoms at different stages when V_{pz} is − 1.6 km/s and V_{px} were used. (a, b) The loading stage. (c, d) The unloading stage. (e, f) The cooling stage.

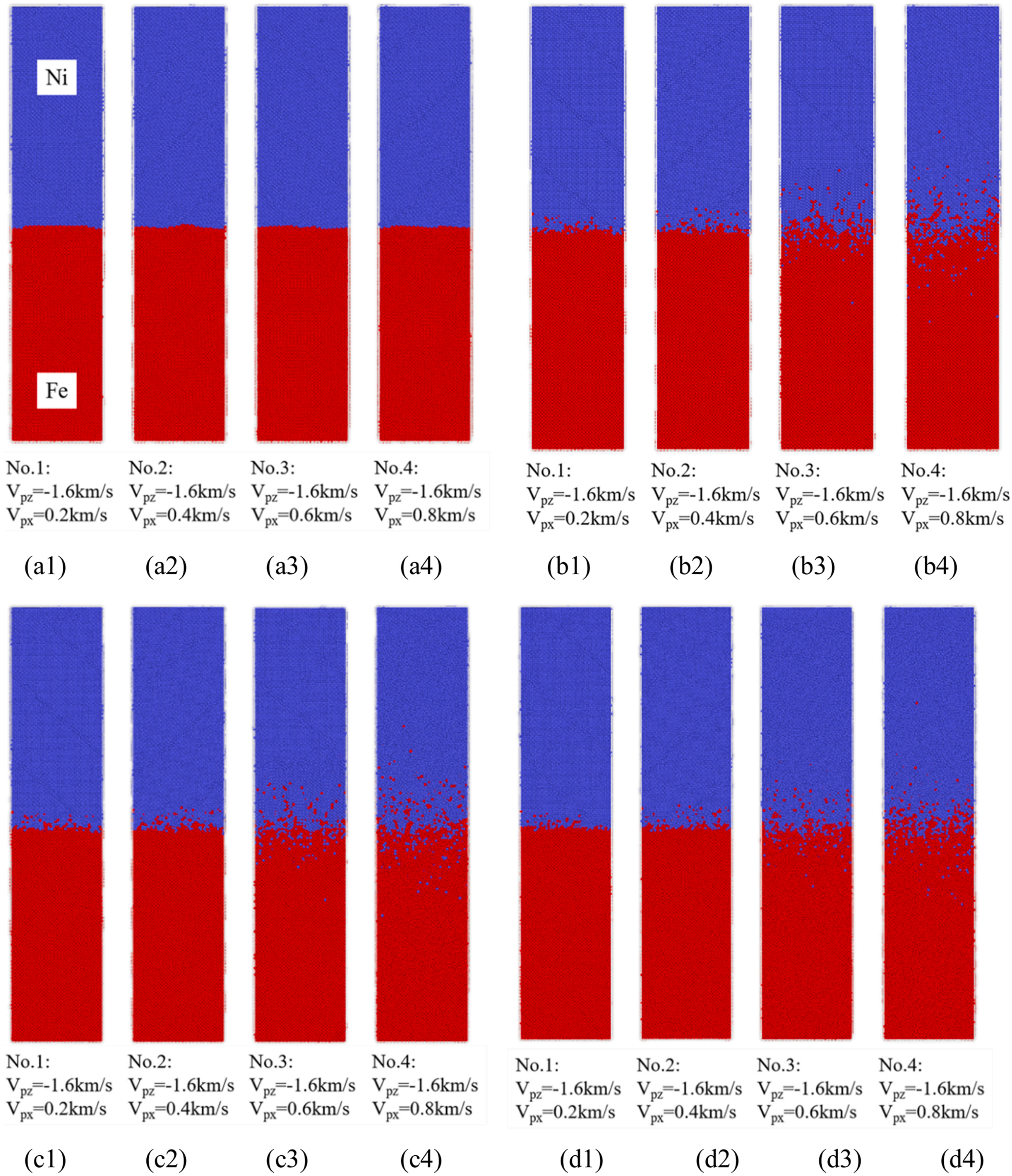

Fig. 7. Configurations of the bonding interface in each stage. (a1-a4) The ending of the loading stage, (b1-b4) The ending of unloading stage, (c1-c4) The beginning of the cooling stage, (d1-d4) The ending of cooling stage.

3. Results and discussion

3.1. Verification of the potential function and model reliability

The applicability of the potential function and the reliability of the model should be the first issues to be discussed in MD simulation. Only if the selected potential function and initial model are stable for the current simulation system, the accuracy of MD simulation could be guaranteed. In this paper, the stability of the atomic structure during the relaxation process is investigated as a method to evaluate the applicability of the potential function and the reliability of the model. In the actual simulation, the structure of the model is not stable and there is a

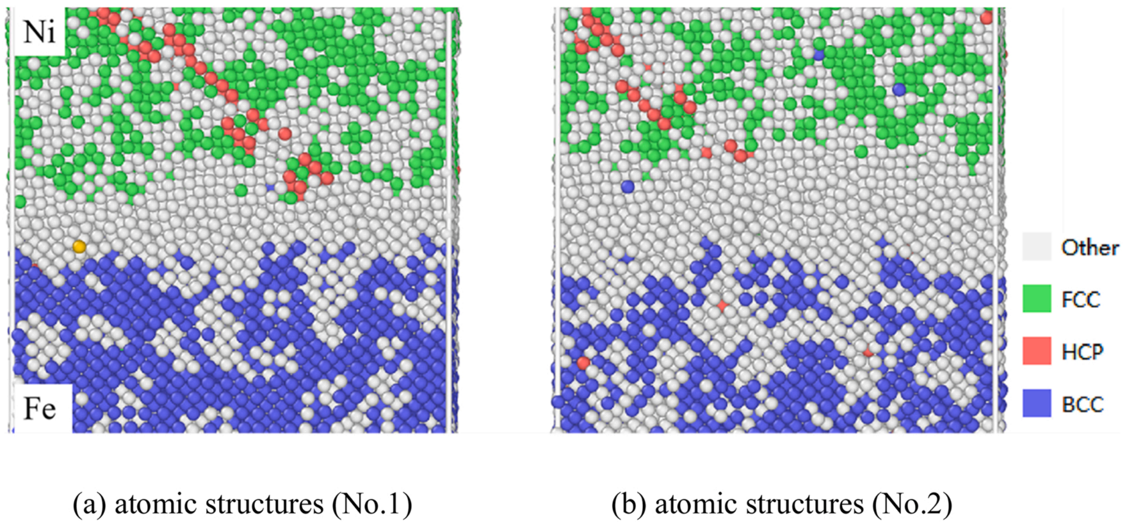

Fig. 8. Atomic structure at the end of the unloading stage with different welding parameters (a, b).

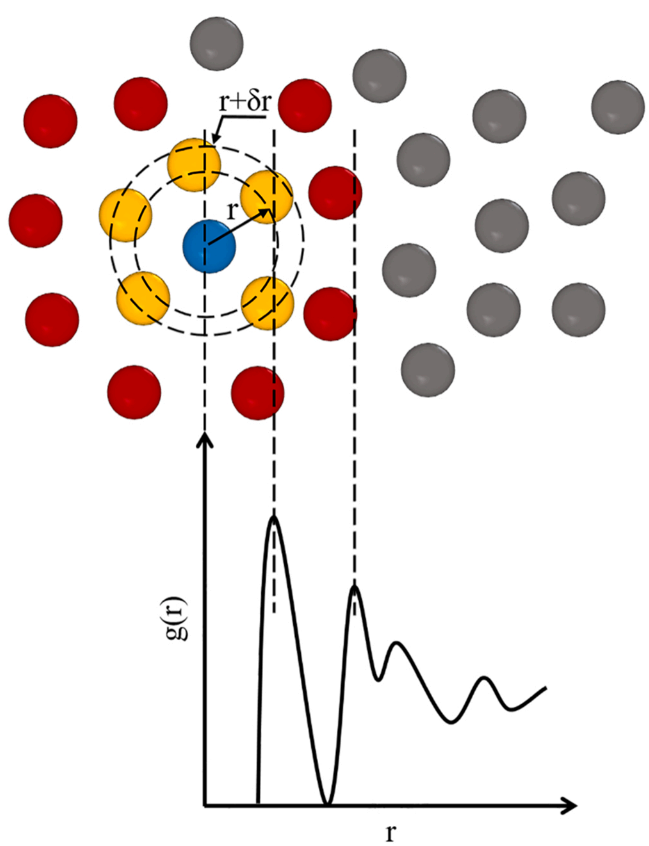

Fig. 9. Schematic diagram of the relationship between the radial distribution function (RDF) and the positions of the atoms.

large prestress, so it is necessary to relax the model to reduce the atomic potential energy. Fig. 4 shows the fractions of atomic crystal structure during the relaxation process. After relaxing the simulation system for about 100 ps, the proportion of BCC crystal structure of Fe changes from 50.2% to 47.5%, while the proportion of FCC crystal structure of Ni changes from 47.7% to 49.7%. After that, the ratios of the atomic structures in the system remain stable. Therefore, it can be considered that the potential function selected in this paper is suitable for the simulation system, and a relatively stable model can be obtained by using this potential function to relax.

3.2. Mean square displacement atomic diffusion behaviour

Fig. 5 shows the temperature and pressure change curves of welding seam system V_{px} = 0.2 km/s, V_{px}= 0.4 km/s, V_{px}= 0.6 km/s, V_{px}= 0.8 km/s in the loading stage of explosive welding. With the collision process of bulks, the initial kinetic energy of the Ni bulk transforms into the internal energy of the whole system, and the temperature and pressure of the simulation system rise dramatically. After about 150 ps, the temperature and pressure of the simulation system maintain the dynamic equilibrium. The equilibrium temperature from MD simulations is 1537 K, 1612 K, 1731 K, 1880 K sequentially, and the equilibrium pressure from MD simulations is 36.3 GPa, 36.8 GPa, 37.1 GPa, 38.3 GPa sequentially.

Fig. 6 shows the MSD curves of the Ni and Fe atoms in the V_{px} = 0.2 km/s, V_{px}= 0.4 km/s, V_{px}= 0.6 km/s, V_{px}= 0.8 km/s weldments at different stages along the Z-axis direction. As shown in Fig. 6(a, b), each MSD curve of Ni and Fe atoms fluctuates dramatically at the beginning of the loading stage due to the collision process. After about 150 ps, each MSD curve of Ni and Fe atoms remains constant. This indicates that Ni and Fe atoms do not diffuse during the loading stage due to the high-temperature and high-pressure environment of the simulated system. According to the reference[9], the melting point of materials increases significantly with the increase of pressure.

At the unloading stage in Fig. 6(c, d), the MSD curves of Fe and Ni atoms in the V_{px}= 0.6 km/s, V_{px}= 0.8 km/s weldments continuously increase with time, indicating that Ni and Fe atoms in the system diffuse into each other. In comparison, the MSD curves of Ni and Fe atoms corresponding to the V_{px} = 0.2 km/s, V_{px}= 0.4 km/s weldments tended to decrease for a short period before becoming constant. At the same time, it was noted that the value of MSD itself was very small at this time. At this time, Ni and Fe atoms in the system do not diffuse. By comparing the differences between weldments V_{px}= 0.6 km/s and V_{px}= 0.8 km/s and weldments V_{px}= 0.6 km/s and V_{px}= 0.8 km/s, the system temperatures of weldments V_{px} = 0.2 km/s and V_{px}= 0.4 km/s are 1537 K and 1612 K, respectively, in the unloading stage, although the weldments are in the environment of 0 external pressure and high temperature. The system temperatures of weldments V_{px}= 0.6 km/s and V_{px}= 0.8 km/s are 1731 K and 1880 K, respectively. The system temperatures of weldments V_{px}= 0.6 km/s and V_{px}= 0.8 km/s are significantly higher than those of weldments V_{px} = 0.2 km/s and V_{px}= 0.4 km/s which indicates that Ni and Fe atoms in the system will have obvious mutual diffusion only when the system temperature reaches a certain height in the unloading stage.

At the beginning of the cooling stage in Fig. 6(e, f), the MSD curves of

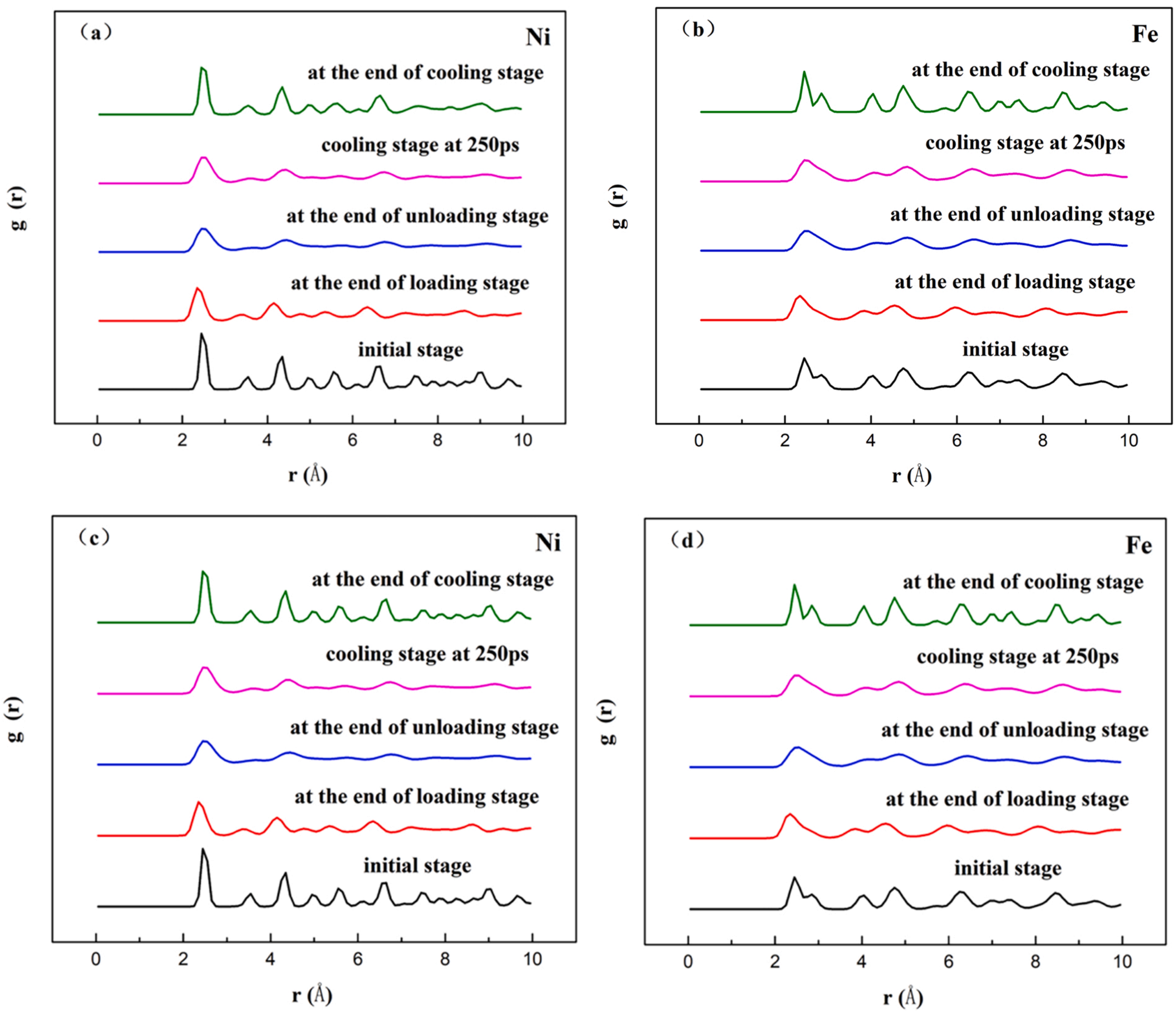

Fig. 10. The RDF results of Ni and Fe at different stages when V_{pz} is fixed at − 1.6 km/s. (a, b) No.1 (V_{px}=0.2 km/s), (c, d) No.2 (V_{px}=0.4 km/s), (e, f) No.3 (V_{px}=0.6 km/s), (g, h) No.4 (V_{px}=0.8 km/s).

Ni and Fe atoms in the V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments remain stable, while the MSD curves of Fe and Ni atoms in the V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments continued to increase with a certain slope at the beginning of the cooling stage, and then kept stable gradually. It can be seen from the slope of MSD curves of Ni and Fe atoms that the slope of MSD curves of each atom in the early cooling stage is less than that in the unloading stage (Fig. 6(c, d)), and that in the V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments Ni and Fe atoms do not diffuse during the cooling phase. But in the V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments, atomic diffusion happens at the beginning of the cooling stage because of the higher temperature environment. When the system temperature drops to a lower level, atomic diffusion gradually stops. According to the temperature and pressure characteristics of the system in the cooling stage, although the four weldments were cooled to room temperature at a certain temperature under 0 external pressure, at the initial cooling stage, weldments V_{px}= 0.6 km/s and V_{px}= 0.8 km/s were still in a high-temperature system, so the system still had an atomic diffusion phenomenon. However, at the later cooling stage, with the decrease of temperature, the atomic diffusion behavior at the interface of weldments V_{px}= 0.6 km/s and V_{px}= 0.8 km/s also stops correspondingly.

3.3. Characterization of atomic diffusion behavior model architecture

The MSD reflects the diffusion of some atoms in the whole system during the welding process, but it can't show the diffusion situation of atoms at a specific moment, and it can't get the diffusion information of atoms at a specified position. To more intuitively describe the atomic diffusion behavior at the interface during explosive welding, the software Ovito was used to observe the system configuration at each stage of explosive welding [37]. Ovito can present the system configuration of atoms at any time point in the explosive welding process. This provides a basis for analyzing the diffusion behavior of interfacial atoms at special time points in the explosive welding process.

Fig. 7 shows the model architecture diagram of four groups of weldments in different stages of explosive welding. As can be seen from Section 3.2 of this paper, the Ni and Fe atoms corresponding to V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments still diffuse towards each other in the early cooling stage. Therefore, Fig. 7 not only shows the system configuration at the end of the loading stage, the end of the unloading stage, and the end of the cooling stage but also observes the system configuration of the 250 ps weldments after the beginning of the cooling stage.

At the end of the loading stage, it can be seen from Fig. 7 (a1- a4) that under the four groups of process parameters, the Ni and Fe atoms only

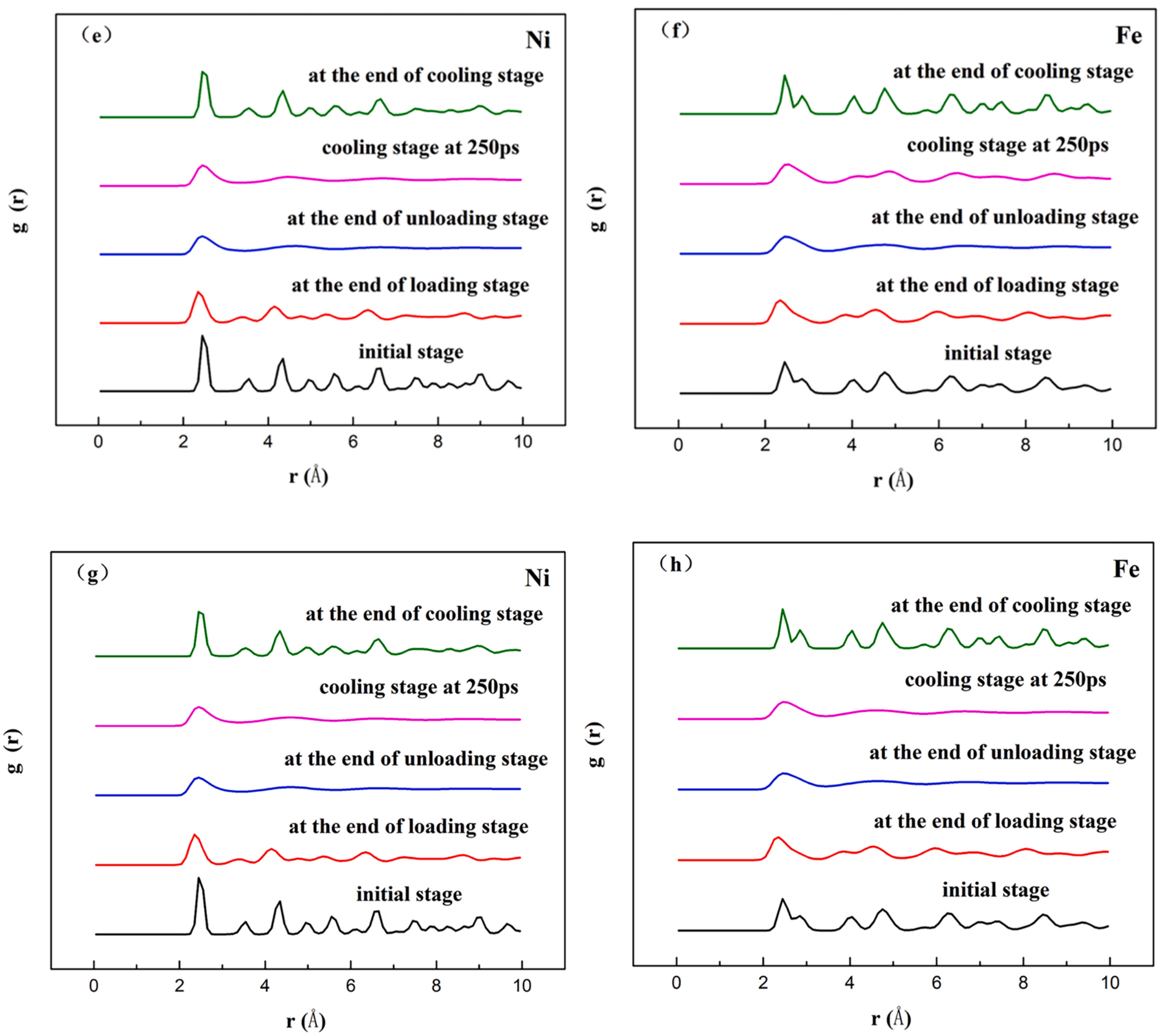

Fig. 10. (*continued*).

Table 3
five weldments of MD simulation under different welding parameters.

No.	V_{pz} (km/s)	V_{px} (km/s)	the type of atoms melted during the welding process
1	-1.6	0.2	neither
2		0.4	neither
3		0.6	Ni、Fe
4		0.8	Ni、Fe
5		0.57	Ni

fluctuate slightly at the interface, and the atoms on both sides do not diffuse to the opposite side, indicating that there is no obvious phenomenon of interatomic diffusion during the whole loading stage.

After entering the unloading stage (Fig. 7 (b1- b4)), while maintaining the temperature at the end of the loading stage, the pressure in the system has been removed. Fig. 7(b1, b2) shows that a small portion of the Fe atoms diffuse to the side of the Ni block at the interface, while a small portion of the Ni atoms diffuse to the opposite side. This phenomenon is inconsistent with the MSD results of V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments in Section 3.2. This is because formula 5 shows that the calculation result of MSD is to all the particles in the system of square difference of displacement, only when the occurrence of obvious diffusion curve of MSD doesn't appear to have an obvious rising trend, and the slope can characterize the diffusion coefficient, and in Fig. 7 (b1, b2), Ni, Fe atom diffusion relative to the number of atoms in the whole system is concerned, there are fewer, Such a small amount of diffusion is averaged over the whole system, which makes it difficult for MSD calculation results to describe the atomic diffusion behavior at the interface of V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments. This is the limitation of the MSD calculation method itself. At the same time, the common neighbor analysis function in Ovito software was used to obtain the crystal structure of atoms near the interface of V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments, as shown in Fig. 8(a, b). By comparing the crystal structure in 8, it can be found that under the current simulated environment of high temperature and 0 external pressure, some Ni atoms near the interface have presented a disordered lattice state at this time, while Fe atoms near the interface have maintained their original BCC crystal structure. In addition, Fe atoms have a smaller atomic radius than Ni atoms. These causes Fe atoms to move in the opposite direction more easily than Ni atoms. Similar phenomena have also been observed in previous work [22].

It can be seen from Fig. 7 (b3, b4) that the atoms corresponding to V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments diffuse obviously near the interface and are distributed symmetrically on both sides of the interface. This phenomenon is consistent with the results of MSD curves of Ni and Fe atoms in the previous unloading stage.

At 250 ps after the beginning of the cooling stage, as shown in Fig. 7

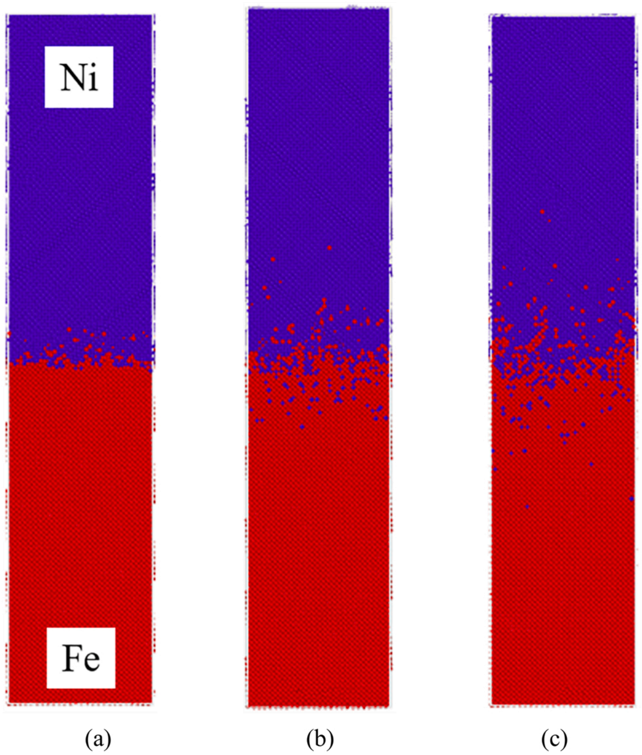

Fig. 11. The final configurations of the bonding interface. (a) No.2 (V_{px}=0.4 km/s), (b) No.5 (V_{px}=0.57 km/s), (c) No.4 (V_{px}=0.8 km/s).

(c1-c4), atoms in V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments still do not diffuse obviously at the interface, while atoms near the interface in V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments diffuse further than those in the unloading phase.

However, at the end of the cooling stage (Fig. 7 (d1-d4)), the four groups of welded parts basically maintain the architecture of the initial cooling stage, and no obvious atomic mutual diffusion is observed. These phenomena are consistent with the previous study [23].

In summary, the atomic diffusion coefficient is mainly affected by the system temperature and pressure in the explosive welding process. And the mutual diffusion mainly occurs in the unloading stage and at the beginning of the cooling stage of explosive welding. The atoms in the simulated system diffuse significantly only when the temperature of the system is high during these phases.

3.4. *Effect of melting on atomic diffusion behavior*

In order to explore the effect of melting on atomic diffusion behavior during the explosive welding process, the radial distribution function (RDF) is used to analyze the atomic structures of each stage [23]. As shown in Fig. 9, if one particle in the system is considered as the reference center, and if the total number of particles divided by the system volume is the average number density of particles (ρ_0), then the local number density of particles within the distance from r to r + δr could be calculated in the same way. The radial distribution function, usually expressed by the term g(r), can be obtained by comparing the average number density of particles with the local number density of particles. The g(r) is a method for calculating the probability of finding a particle at a distance of r from a given reference center[38]. The formula for g(r) is given by:

$$g(r) = \frac{n(r)}{V\rho_0} \approx \frac{n(r)}{4\pi r^2 \rho_0 \delta r} \quad (7)$$

Where *r* is the distance from the reference particle, *δr* is the thickness of the spherical shell, n(r) and V are the local number density of particles and the volume within the distance from r to r + δr, ρ_0 is the average number density of total particles. Because the liquid exhibits both short-range order and long-range disorder, the g(r) will have a sharp peak in the short-range and will remain constant in the long-range. Thus, it can be used to investigate whether melting occurs during the explosive welding process. Because of its long-range order properties, it can be used to investigate atomic structure changes during the explosive welding process by observing the distribution of g(r) peaks.

Fig. 10 shows the RDF results of Ni and Fe at different stages. In the loading stage, as shown by the solid red line in Fig. 10 (a-h). Because the system was subjected to a collision and high-pressure environment during the loading stage, the position of the first peak of g(r) shifts to the left from its initial position, and no evidence of atomic melting is found for the four weldments. When it comes to the unloading stage, as shown by the solid blue line in Fig. 10 (a-h). The Ni and Fe atoms in the system corresponding to the V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments were still maintained in Fig. 10 (a-d) according to the principle of the g (r) maintaining at 1 at the long-range for liquid. However, the Fe and Ni atoms in the system corresponding to the V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments have melted, as shown in Fig. 10 (e-h). The same phenomenon was observed in Reference[23]. Although the four weldments were all under the high temperature and zero external pressure environment during the unloading stage, the system melted only in the V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments because of the higher environmental temperature. At the beginning of the cooling stage (250 ps after the start of the cooling stage), as shown by the solid pink line in Fig. 10 (a-h), the Fe and Ni atoms maintain the solid structure in the V_{px} = 0.2 km/s and V_{px}= 0.4 km/s weldments; while the Fe and Ni atoms still maintain the liquid structure in the V_{px}= 0.6 km/s and V_{px}= 0.8 km/s weldments because of the high temperature and super-cooled phenomena caused by the fast cooling rate. After the cooling phase was completed, the atoms of the four welding parts were all solid structures, and the RDF results were similar to the initial structure. In combination with the conclusion in Section 3.3, it can be seen that atoms in the inter-diffusion interface mainly occur in the unloading stage and at the beginning of the cooling stage. If the system does not melt at this time, only a small number of atoms can diffuse to the opposite interface. Melting occurs only at these stages for atoms at the bonding interface to diffuse appreciably.

The above simulation results show that during the welding process, no melting occurred at either end of the welding seam system of the V_{px} = 0.2 km/s and V_{px}= 0.4 km/s. V_{px}= 0.6 km/s and V_{px}= 0.8 km/s both melted during the welding process. Reference[23] considers that melting occurs at the unloading stage when the temperature is higher than the melting point. Therefore, when the V_{pz} is fixed at − 1.6 km/s and the V_{px} is in the range of 0.4 ~ 0.6 km/s, during the welding process, one kind of atom will melt and the other kind of atom will not. After several simulation attempts, the simulation results of Ni atomic melting in the welding process were obtained when V_{pz} is − 1.6 km/s and V_{px} is 0.57 km/s (defined as No.5 weldment). The five weldments simulated by MD were summarized in Table 3.

Fig. 11 shows the final configuration of the bonding interface in the V_{px} = 0.4 km/s, V_{px}= 0.57 km/s and V_{px}= 0.8 km/s weldments sequentially. As shown in Fig. 11 (a), although the V_{px} = 0.4 km/s weldment did not melt at both ends of the system during the welding process, a small number of Fe atoms were diffused to the opposite side of the interface, while Ni atoms had no obvious diffusion phenomenon on the opposite side of the interface. At this time, only a small amount of solid-solid diffusion occurred at the explosive welding interface. The reason for solid-solid diffusion can be explained by the Kirkendall effect. Ni melting point is lower than the melting point of Fe, which means that the bonding effects between Ni and Ni atoms in the lattice in are easier to break, and the bonding effects between Fe atoms break require more energy, by contrast, it is easier to produce Ni lattice vacancy or other defects, these will be good for Fe atoms of space and defect formation in

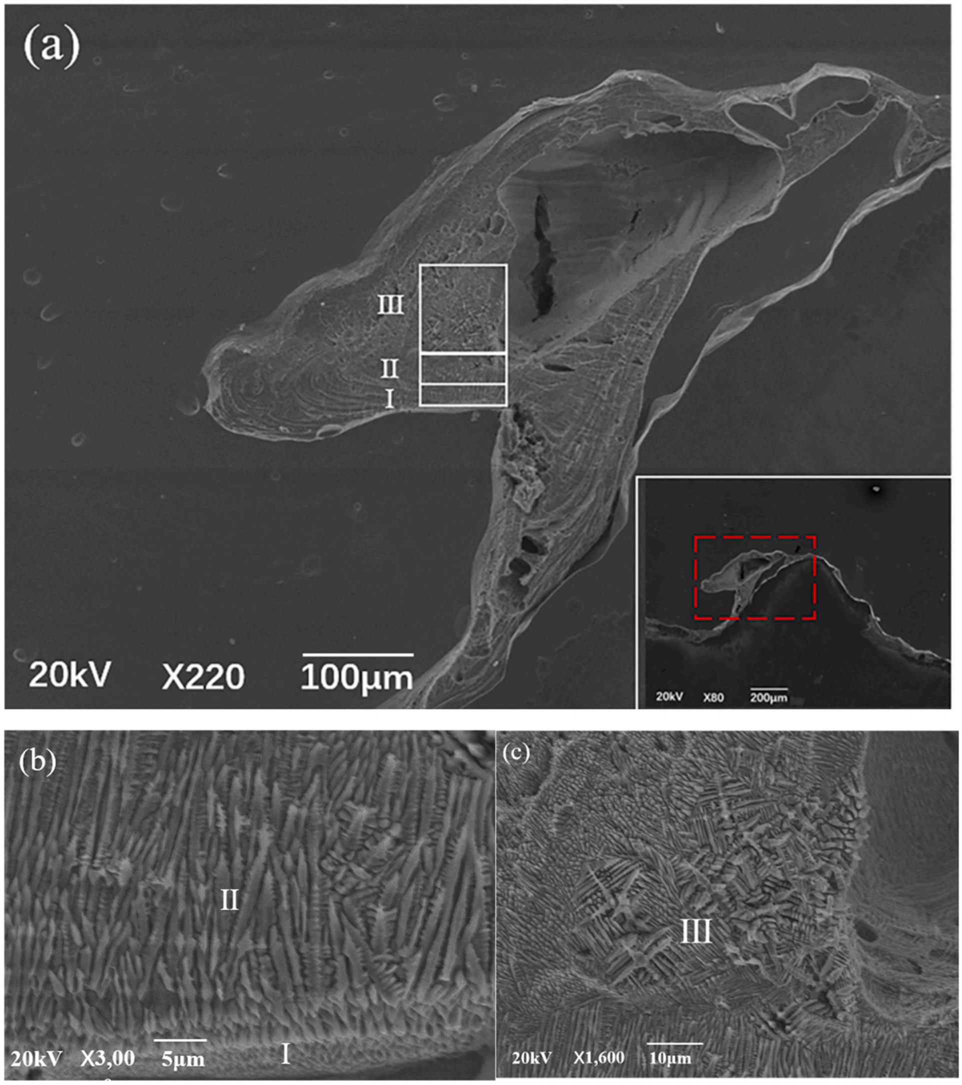

Fig. 12. SEM image of microstructure across the wavy interface.

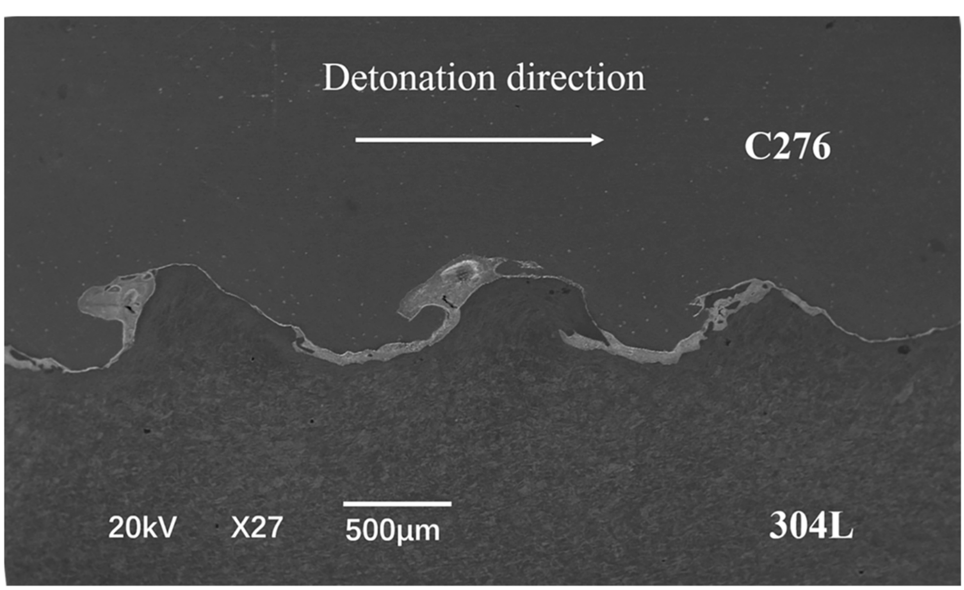

Fig. 13. Local amplified SEM images of the vortex on the right side of the wave peak.

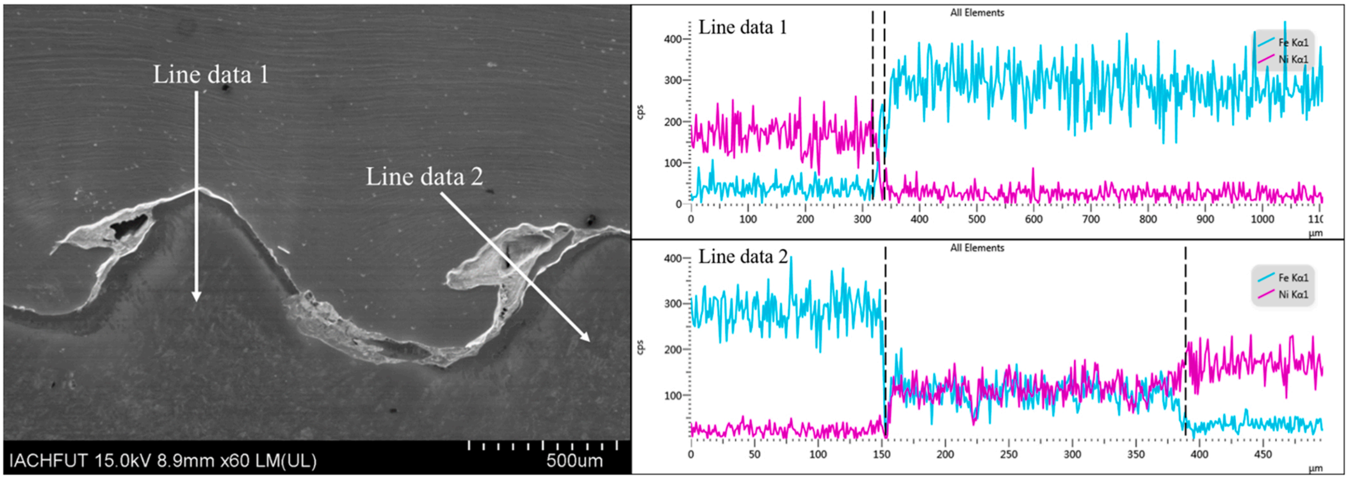

Fig. 14. Main element distribution of the C276/304 L composite by EDS line scan.

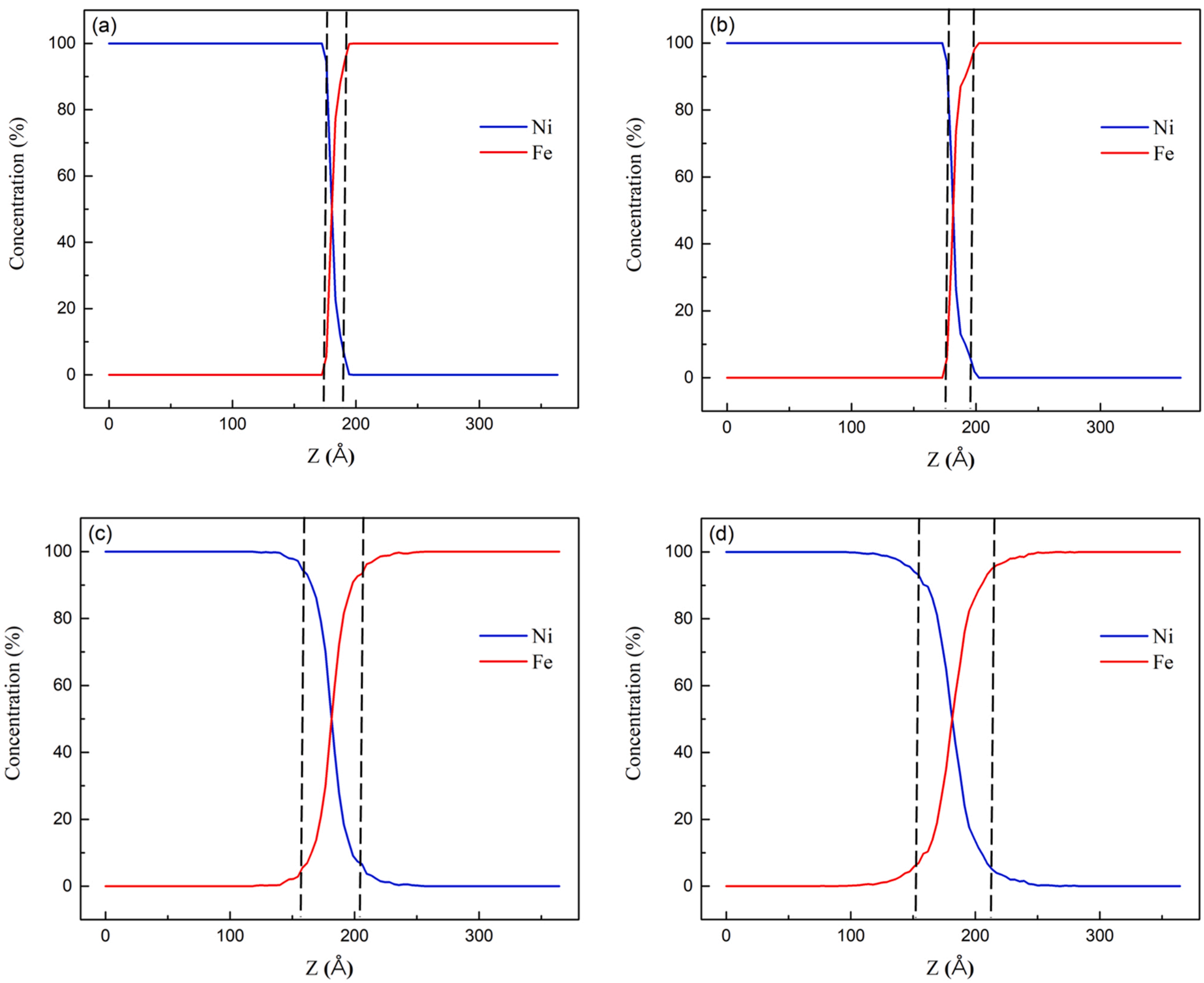

Fig. 15. Main element distribution of MD simulation along the Z-axis when V_{pz} is fixed at − 1.6 km/s. (a) No.1 (V_{px}=0.2 km/s), (b) No.2 (V_{px}=0.4 km/s), (c) No.3 (V_{px}=0.6 km/s), (d) No.4 (V_{px}=0.8 km/s).

the Ni lattice diffusion, The number of Fe atoms diffusing into Ni is more than that of Ni atoms diffusing into Fe. The excess Fe atoms cause the lattice of Ni to expand, while the lattice contract occurs where the Ni atoms decrease, and less diffusion occurs at this time. In Fig. 11 (b), although only Ni atoms melt and Fe atoms do not melt, Ni and Fe atoms in the system still have an obvious mutual diffusion phenomenon on both sides of the interface. In a collision, atoms on the contact surface transfer energy to several atoms of different kinds and directions,

Table 4
The MD simulation under different welding parameters.

No.	V_{pz} (km/s)	V_{px} (km/s)	the type of atoms melted during the welding process	The thickness of diffusion layer (Å)
1	-1.6	0.2	neither	14
2		0.4	neither	20
3		0.6	Fe、Ni	49
4		0.8	Fe、Ni	61
5	-1.8	0.2	Fe、Ni	63
6		0.4	Fe、Ni	67
7		0.6	Fe、Ni	75
8		0.8	Fe、Ni	85
9	-2.0	0.2	Fe、Ni	86
10		0.4	Fe、Ni	99
11		0.6	Fe、Ni	118
12		0.8	Fe、Ni	132
13	-2.2	0.2	Fe、Ni	110
14		0.4	Fe、Ni	125
15		0.6	Fe、Ni	141
16		0.8	Fe、Ni	150

causing more atoms to displace. Atoms transfer energy along the depth direction of the plate. In the vertical direction, due to the principle of momentum conservation, atoms will also collide with a cluster of atoms, resulting in the generation of vacancy and gap atoms. After the collision, the energy of all atoms is lower than the ionization threshold energy, after which the ionized atoms gradually decrease and become stable. As shown in Fig. 11 (c), during the welding process, Ni and Fe atoms at both ends of the interface of the weldment $V_{px} = 0.8$ km/s system have melted, and Ni and Fe atoms have diffused on both sides of the interface. In this case, the interface diffusion should be liquid-liquid diffusion, and the degree of diffusion is greater than that of weldment $V_{px} = 0.57$ km/s. In the welding process of the $V_{px} = 0.4$ km/s, when neither of the two atoms can be fused during the welding process, it can be seen that only a few Fe atoms were diffused to the side of the Ni bulk near the interface. In the welding process of the $V_{px} = 0.57$ km/s, there was part of the Fe and Ni atoms near the interface that could inter-diffuse when only the Ni atoms melted during the welding process. In the welding process of the $V_{px} = 0.8$ km/s, two types of atoms could melt, and the diffusion behavior of atoms on both sides of the interface is greater than that of the $V_{px} = 0.57$ km/s weldment. From these results it can be concluded: when the atoms at both ends of the weldment do not melt during the welding process, the atomic diffusion on both sides of the interface is not obvious. When one end of the weldment melts, the other end does not melt, or even both ends of the weldment melt, there will be an obvious mutual atomic diffusion phenomenon.

According to the above results, whether atoms at both ends of the weldment melt in the explosive welding process will directly affect the degree of atomic diffusion on both sides of the interface, which explains why there has been controversy on whether the explosive welding interface diffusion occurs in the previous studies of scholars. Previous studies have mostly focused on experimental studies. Due to the limitations of experiments, it is difficult to demonstrate the three kinds of diffusion at the explosive welding interface: solid-solid diffusion, solid-liquid diffusion, and liquid-liquid diffusion. When solid-solid diffusion occurs at the explosive welding interface under certain process parameters, the experimental results show that there is little or no diffusion at the explosive welding interface. Under the experimental conditions, solid-liquid diffusion and liquid-liquid diffusion are even more difficult to distinguish, and the unified conclusion can be drawn that the explosive welding interface appears to have obvious diffusion.

Due to the limitation of the time scale of the MD simulation, the duration of the unloading stage in this simulation is 1000 ps and the cooling rate in the cooling stage in this paper is set at 10^{12} K/s, and the inter-diffusion of interfacial atoms mainly occurs during the unloading stage and the beginning of the cooling stage. In the actual welding process, the unloading stage is usually about 5–10 μs [9], which is much longer than that of MD simulation. At the same time, the cooling rate of the temperature near the interface in the welding process is about 105–10^9 K/s [39–41]. In actual welding, the effect of melting on the diffusion behavior of atoms near the interface is more significant than in MD simulation. Therefore, in explosive welding, atomic diffusion should occur at the interface of explosive welding no matter whether the atoms at both ends melt or not. The melting of atoms at the interface will accelerate the diffusion of elements near the interface.

To explore the effect of the melting on the atomic diffusion behavior near the interface in the actual welding process. This article takes Hastelloy/stainless steel explosive welded plate as the research object to investigate the interfacial morphology and microstructure.

Fig. 12 shows the local amplified SEM (JSM-64901 V) images of the vortex on the left side of the wave peak of the wavy interface. It can be found that the local microstructure of the vortex can be roughly divided into three regions, which are the region of equiaxed dendritic grains at the edge of the vortex (I), the region of columnar dendritic grains (II) and the mixed region of the equiaxed dendritic grains and columnar crystals at the center of the vortex (III). The proof of metal melting in the presence of dendritic grains. Therefore, it can be determined that these areas have melted during the welding process. As shown in Fig. 13, there is a molten metal layer in the bonding region of the C276/304 L composite material, and it presents a wavy interface. There are obvious

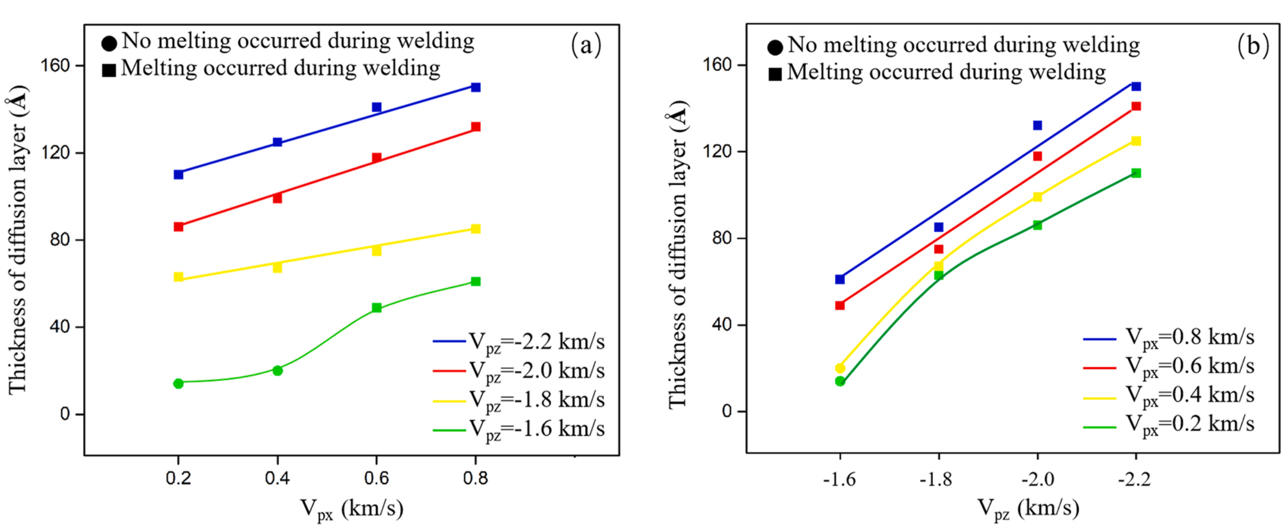

Fig. 16. The relationship between the welding parameters and atomic diffusion behaviour when the V_{pz} are fixed (a) and the V_{px} are fixed (b).

differences in the thickness of the diffusion layer at different positions of the wavy interface.

Fig. 14 shows the main element distribution of the C276/304 L composite near the interface by EDS line scan. The scan directions are perpendicular to the welding interface (line data 1) and the vortex (line data 2) from the flyer plate to the base plate. It can be seen that the thicknesses of the diffusion layer at the wave peak and the vortex zone are about 20 μm and 230 μm respectively. At the same time, the distribution of the main elements in the vortex area tends to be uniform, and it can be determined that both sides of the C276/304 L composite near the interface were melted during the welding process. However, in previous explosive welding works, such as the Cu/steel explosive welded plate of Durgutlu et al. [19], the 316 L(N)/CuCrZr explosive welded plate of Ma R et al. [42], and the Gr5/SS304 explosive welded plate of Zhao H et al. [43], no signs of melting were found near the interface, and these weldments did not have an obvious diffusion layer. Based on these results, it can be concluded that the interfacial atoms can significantly inter-diffuse and the weldment can form an obvious diffusion layer near the interface only when the welded material near the interface melts during the explosive welding process. This conclusion can reasonably explain the phenomena where some researchers found that the interfacial atoms can diffuse into the opposite side and form an obvious diffusion layer during the explosive welding process [18], while other researchers found that the interfacial atoms can hardly form an obvious diffusion layer [19]. Besides, it can be seen from this conclusion that whether the explosive welding experiment melts and forms an obvious diffusion layer depends on the melting point and welding parameters of the welding material. However, in actual experiments, the thin layer of the diffusion layer near the interface limits the formation of intermetallic compounds [44], and some brittle intermetallic compounds significantly reduce the bonding strength of the welded joint [45–47]. Therefore, the thickness of the diffusion layer of the weldment should be appropriate and it requires a reasonable selection of welding parameters.

3.5. Effect of welding parameters on atomic diffusion behavior

To quantify the atomic diffusion behavior near the interface, the Python script is used to extract the atomic coordinate data from the MD simulation results. Fig. 15 shows the distribution of atomic concentration in the final system of the $V_{px} = 0.2$ km/s, $V_{px}= 0.4$ km/s, $V_{px}= 0.6$ km/s, $V_{px}= 0.8$ km/s weldments along the Z-axis. By defining the interfacial region where the concentration of Ni and Fe atoms both exceeds 5% as the diffusion layer, the z-length of the diffusion layer can be calculated. It can be seen from Fig. 15 that the thickness of the diffusion layer of the $V_{px} = 0.2$ km/s, $V_{px}= 0.4$ km/s, $V_{px}= 0.6$ km/s, $V_{px}= 0.8$ km/s weldments is about 14, 20, 49 and 61 Å. In addition, this method can be used to investigate the thickness of the diffusion layer under other welding parameters.

To explore the influence of welding parameters on atomic diffusion behavior during explosive welding, 16 sets of V_{pz}-V_{px} welding parameters are used to simulate the explosive welding process, as shown in Table 4. Also, the RDF results are used to investigate the type of atoms melted during the welding process, and the Python script is used to investigate the thickness of the diffusion layer. All of the MD simulation results are summarized in Table 4. Fig. 16 shows the relationship between the welding parameters and atomic diffusion behavior when the V_{pz} or V_{px} are fixed. At the same time, by comparing the slopes of the curves in Fig. 16 (a, b), it can be seen that under the premise of an increment of 0.2 km/s, the contribution of increasing the absolute value of V_{pz} to increasing the thickness of the diffusion layer is significantly higher than that of increasing the absolute value of V_{px} to increasing the thickness of the diffusion layer. Furthermore, as shown in Fig. 16(a), the thickness of the diffusion layer increases to varying degrees as the absolute value of V_{px} increases. When $V_{pz} = -1.6$ km/s, because $V_{px} = 0.2$ km/s and 0.4 km/s, the system did not melt during welding (in Fig. 16, the failure of melting was indicated by a circle, while the melting was indicated by a square), the thickness increment of the diffusion layer was small. When V_{px} increases to 0.6 km/s, the thickness of the diffusion layer increases significantly due to the melting during welding and keeps linear growth after that. A similar phenomenon can be observed in Fig. 16 (b).

In the process of explosive welding, the formation of the jet at the bonding point and the occurrence of melting contribute to the further diffusion of atoms at the interface, thus helping the atoms on both sides of the interface form a good metallurgical bond. However, excessive melting of the interface in the welding process will lead to the formation of a large number of casting defects on the interface, resulting in a reduction in the bonding strength of the composite. In addition, the short diffusion area at the interface can limit the formation of brittle and hard intermetallic compounds. Therefore, in the actual test, it is required that there is a certain amount of mutual diffusion of atoms at the interface, and it is not expected that excessive melting occurs at the interface during the welding process. This requires technicians to select reasonable welding process parameters.

4. Conclusions

The diffusion behavior of the atoms of the main elements (Ni, Fe) at the interface during the explosive welding process was studied by using molecular dynamics theory and experimental research. The main conclusions are as follows:

1. At the beginning of the loading stage, there is no diffusion phenomenon at this time. In the unloading stage, the obvious mutual diffusion occurs only when the system temperature reaches a certain height. In the cooling stage, the diffusion at the interface stops gradually with the decrease in temperature.
2. In explosive welding, atomic diffusion will occur at the interface atoms. If there is no melting at both ends of the system, only a few atoms at the interface will diffuse to the opposite side of the interface, which is called solid-solid diffusion. During the unloading stage and the initial cooling stage, the system melts, and atoms at the interface diffuse with each other, which is liquid-solid diffusion or liquid-liquid diffusion.
3. The melting of atoms at the interface accelerates the diffusion of atoms near the interface. In the experimental study, the interface of the Hastelloy alloy/stainless steel explosion-welded composite plate presents typical wavy interface characteristics, and the typical as-cast structure appears in the vortex area, which proves that the base metal at the interface melted during the welding process. EDS results show that there is an obvious diffusion phenomenon at the interface.
4. Within the range of process parameters, increasing the absolute value of the transverse velocity component V_{px} and the longitudinal velocity component V_{pz} will increase the thickness of the diffusion layer. Increasing the absolute value of the longitudinal velocity component V_{pz} of the cladding will significantly improve the thickness of the diffusion layer.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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